

TraceFinder: Forensic Scanner Identification

1. Introduction

TraceFinder is a forensic project aimed at identifying the exact source scanner device used to digitize a document. Every scanner leaves a unique, nearly invisible signature—a pattern of noise and texture—that acts like a device "fingerprint". By applying image processing and machine learning techniques, this project aims to train a classifier to learn and recognize these specific scanner characteristics. The final goal is a system that can accurately recognize the scanner model based only on the scanned image

Milestone 1 – Dataset Collection & Preprocessing

2.1. Introduction

This milestone focused on understanding the raw scanned-document dataset, analyzing basic properties, and preparing a consistent preprocessing pipeline. The ultimate goal was to clean, normalize, and structure the dataset so that it is ready for feature extraction and model training in later stages.

2.2 Scanner Models Present in the Dataset

The dataset includes 11 scanner models, each contributing Flatfield, Official, and Wikipedia image samples:

The full project dataset, containing samples from multiple scanner devices, was categorized into three main labels based on content type:

- *Wikipedia*
- *Official*
- *Flatfield*

2.3. Dataset Exploration and Key Observations

Initial analysis was performed on the full dataset, which comprised a Total size of 4,590 images. Key observations revealed significant variability and challenges for the machine learning model. Specifically, the data was highly imbalanced, with 2,368 images in the 'Wikipedia' category but only 22 in the 'Flatfield' category. Resolution was also not uniform, showing two dominant native sizes: a 150 DPI range (approximately 1240 x 1754 or 1236 x 1754) and a 300 DPI range (approximately 2480 x 3508 or 2400 x 3500). Furthermore, all 4,590 images were originally in the RGB color mode. Regarding content, Flatfield images showed very low entropy, while Official and Wikipedia images

showed higher entropy, containing text. Finally, data indexing was successfully completed by creating manifest files (e.g., flatfield_preprocessing.csv) that linked the full file path to the scanner ID and a numeric label ID.

2.4. Preprocessing Pipeline

A consistent preprocessing strategy was designed and applied to all images to ensure the final output is standardized and focused on noise-based features. The overall target size was set to 512 x 512. This pipeline began with Grayscale Conversion, which simplifies processing and focuses analysis on structural and noise-based features by converting all images to 8-bit grayscale (1 channel). Next, Resizing was performed to create a uniform input size required for the classifier by adjusting all images to the target dimension of 512 x 512. The final step was Normalization, which converts pixel values to float and scales the range from 0 to 255 to the range of 0 to 1, ensuring the data range is optimized for model convergence.

Visualization and Verification

Random samples from the 'Official' and 'Wikipedia' categories were visualized to verify the transformation pipeline. For instance, an Official Sample that started as a high-resolution, color document (e.g., 3508 x 2480) was successfully transformed into the 512 x 512 grayscale, standardized input required for the model. Similarly, a Wikipedia Sample with a different starting resolution (e.g., 1752 x 1240) was also converted into the required 512 x 512 grayscale, standardized input. This verification confirms that the preprocessing pipeline consistently handles all input variability.

Conclusion

Milestone 1 is complete. The full dataset has been successfully analyzed, indexed, and transformed. The images now exist as standardized NumPy arrays of shape (N, 512, 512, 1), meaning they are uniform in size, grayscale, and normalized. The dataset is now prepared and ready for the next phase: Milestone 2, Feature Engineering and Baseline Modeling.
